In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2017

CONCEPT RECAPITULATION TEST - III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 10)** contains 10 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 – 10)** contains 10 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

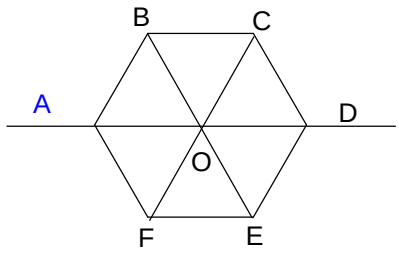
PART – I

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

1. 12 identical rods of length ℓ area cross section area A and thermal conductivity K are connected to form a regular hexagon ABCDEF with centre as O as shown in figure. If heat energy entering at point A at rate $\frac{dQ}{dt} = P$ and leaving from point D. Temperature at end A is T then



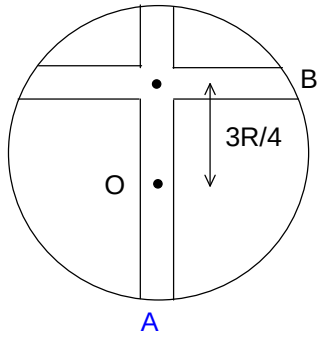
(A) temp at D is $\left(T - \frac{12\ell P}{KA}\right)$

(B) temp at O is $\left(T - \frac{2\ell P}{5KA}\right)$

(C) temp at B is $\left(T - \frac{3\ell P}{10KA}\right)$

(D) temp at D is $\left(T - \frac{4\ell P}{5KA}\right)$

2. Two tunnels are dug through the earth as shown in figure. O is the centre of earth. A particle B is released from rest from position shown and travels through the tunnel. At the same time particle from A is projected toward the centre O with velocity V such that both A and B collide before they could change their direction of motion. Then the velocity V given to A is

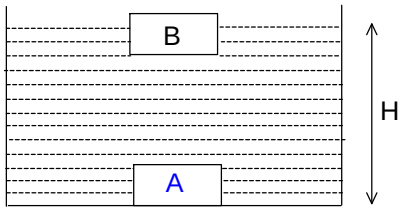


(A) $\frac{3}{4}\sqrt{\frac{GM}{R}}$ (B) $\frac{5}{12}\sqrt{\frac{GM}{R}}$

(C) $\frac{4}{9}\sqrt{\frac{GM}{R}}$ (D) $\frac{4}{5}\sqrt{\frac{GM}{R}}$

Space for rough work

3. Figure shows block B partially immersed in liquid and block A sunk completely and settled at bottom of the container. Area of the cross section of the container is always constant and equal to a. Assume that $d_A > d_L > d_B$ α and γ represent the coefficient of linear and volume expansion respectively. Choose correct statement.



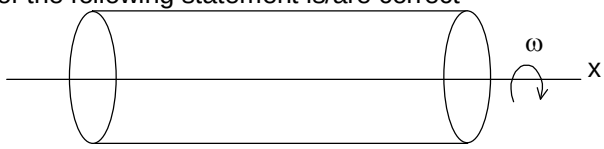
(A) if $\alpha_\beta \neq 0$ and $\alpha_A = \gamma_L = 0$ then on increasing the temp of the system level of liquid H will increase

(B) if $\alpha_\beta \neq 0$ $\alpha_A = \gamma_L = 0$ then increasing the temp of system buoyant force on B increase

(C) if $3\alpha_\beta > \gamma_L$ then on decreasing the temp of the system the fraction of volume of B immersed will increase

(D) if $\alpha_A \neq 0$ $\alpha_\beta = \gamma_L = 0$ then on increasing the temp of the system buoyant force on A will increase

4. A long thin wooden cylindrical shell of radius R carries uniform surface charge density ρ. It is rotating about its axis with angular velocity ω. Which increases slowly with time t as $\omega = kt$ where k is constant. Which of the following statement is/are correct



(A) both the magnetic and the electric field are uniform and constant

(B) the magnetic field is uniform but not constant and the electric field is constant but not uniform

(C) the magnetic field is constant but not uniform electric field is uniform but not constant

(D) total energy U stored in both types of field varies with time as $U = a + bt^2$ where and b are positive constant

5. A metal plate when exposed to light of wavelength λ photoelectrons are ejected when a retarding electric field of intensity E is applied none of the photoelectrons can move away from the plate farther than a certain distance d. Which of the following correct expression for the threshold wavelength λ.

(A) $\lambda_{th} = \dfrac{hc}{eEd}$

(B) $\lambda_{th} = \lambda - \dfrac{hc}{eEd}$

(C) $\lambda_{th} = \left(\dfrac{1}{\lambda} - \dfrac{eEd}{hc}\right)^{-1}$

(D) none of these

Space for rough work

6. In the circuit are shown two parallel plate capacitor of capacitance C and 2C consisting of plates of identical dimensions having separation between two plates 2d & d respectively has shown in figure. They are connected with an ideal battery of terminal voltage as shown in figure – I. Now capacitor smaller in volume is completely inserted into the other capacitor larger in volume and kept in symmetric position and then connection polarities of the plates are reversed as shown in figure – II. Identify correct statements.

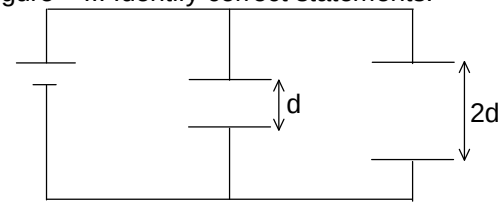


Figure - I

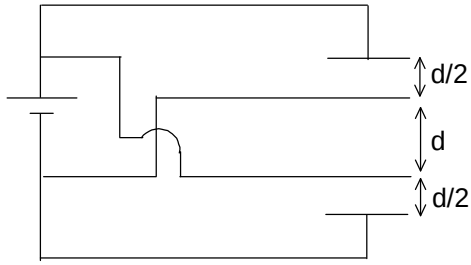


Figure - II

(A) charge flows through battery is 10 CV and battery absorbed energy
(B) charge flows through battery is 10 CV and battery delivers energy
(C) charge flows through battery is 7 CV and battery absorbs energy
(D) charge flows through battery is 7 CV and battery delivers energy

7. Two mirrors M_1 and M_2 make an angle ϕ with line AB. A point source S is kept at a distance r from the point of intersection of mirror along line AB. A small hemispherical annular mask is kept close to S so that no ray emanating from S directly reaches the screen kept at distance b from the source. Interference pattern of rays reflected from M_1 and M_2 is observed on screen. Find the fringe width of this interference pattern λ = wavelength of light being emitted by source.

(A) $\dfrac{\lambda\left(b+2r\sin^2\phi\right)}{2r\sin 2\phi}$

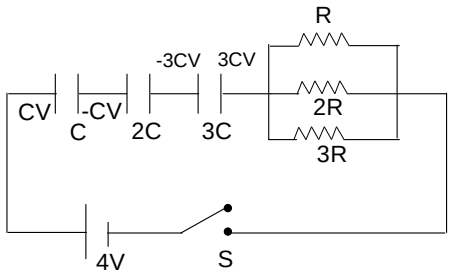
(B) $\dfrac{\lambda\left(b+2r\cos 2\phi\right)}{2r\sin 2\phi}$

(C) $\dfrac{\lambda\left(b+r\cos\phi\right)}{2r\sin 2\phi}$

(D) $\dfrac{\lambda\left(b+2r\cos 2\phi\right)}{r\sin 2\phi}$

Space for rough work

8. Three capacitors of capacitance C, 2C and 3C and three resistors R, 2R and 3R are connected with a battery and switch S as shown in figure. When the switch is open charge on capacitors are shown. Switch S is closed at t = 0. Time constant for the circuit is



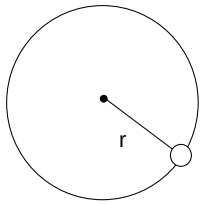
(A) RC

(B) $\frac{6RC}{11}$

(C) $\frac{36RC}{121}$

(D) none of these

9. A small collar of mass m is given an initial velocity of magnitude V_0 on the horizontal circular track fabricated from a slender rod. If co-efficient of friction is μ_k. Determine distance travelled by collar before comes to rest (Recognize that the friction force depends on the net normal force).



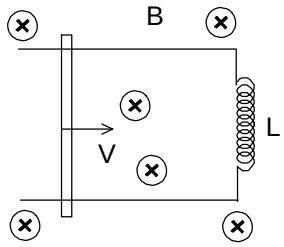
(A) $\frac{r}{\mu_k}\ln\left[\frac{V_0^2+\sqrt{V_0^4+r^2g^2}}{rg}\right]$

(B) $\frac{r}{4\mu_k}\ln\left[\frac{V_0^2+\sqrt{V_0^4+r^2g^2}}{rg}\right]$

(C) $\frac{2r}{\mu_k}\ln\left[\frac{V_0^2+\sqrt{V_0^4+r^2g^2}}{rg}\right]$

(D) $\frac{r}{2\mu_k}\ln\left[\frac{V_0^2+\sqrt{V_0^4+r^2g^2}}{rg}\right]$

10. Two parallel resistanceless rail are connected by an inductor of inductance L at one end as shown in figure. A magnetic field exist in space which is $\perp_r$ to plane of the rails. Now a conductor of length ℓ and m is placed transverse on rail and given an impulse J towards rightward direction. Select correct option



(A) velocity of conductor is half of initial velocity after displacement of conductor $d=\sqrt{\frac{3J^2L}{4B^2\ell^2m}}$

(B) velocity of conductor is half of initial velocity after displacement of the conductor $d=\sqrt{\frac{3J^2L}{B^2L^2m}}$

(C) current flowing through the inductor at the instant where velocity is half of initial is $I=\sqrt{\frac{3J^2}{4Lm}}$

(D) current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $I=\sqrt{\frac{3J^2}{mL}}$

Space for rough work

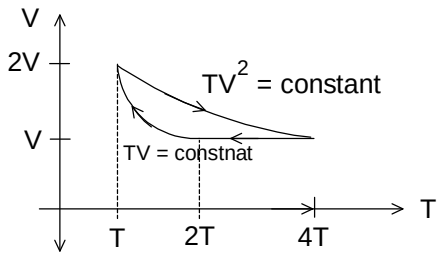
SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

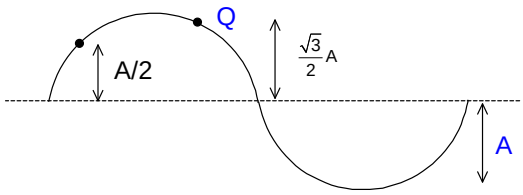
1. A uniform disk rotating about its vertical axis is placed on a horizontal floor divided in two halves. The co-efficient of friction are $\mu_1 = 0.78$ and $\mu_2 = 0.67$ between the disk and right and left halves respectively. Consider the line dividing two half as y-axis and point where centre of disk lies on origin of co-ordinate system. Acceleration of the centre of disk immediately after it is placed on floor is $\frac{K}{20} ms^{-2}$. $g = 10\ ms^{-2}$, $\pi = \frac{22}{7}$.

2. Figure shows VT diagram for helium gas in a cyclic process. Then ratio of maximum and minimum pressure is



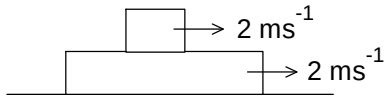
3. In a thin vertical U tube of very long arms is m_1 mass of mercury. Period of small amplitude oscillation of mercury is $T_1 = 2S$. If in one arm $m_2 = 100$ g of water is poured the period of small oscillation becomes $T_2 = 3S$ then mass $m_1 = 16$ K gm. Find K …

4. Figure show the snapshot of the transverse sinusoidal wave on a string at time t = 0 sec. Frequency of wave is $\frac{125}{6} Hz$. T_1 is the time at which the line joining P and Q become horizontal for first time (after t = 0) if wave is travelling toward +ve x direction and T_2 is the time at which the line P and Q becomes horizontal for first time (after t = 0) if wave is travelling in – ve x direction then $\frac{T_1}{T_2} = \frac{22}{K}$. Find K …

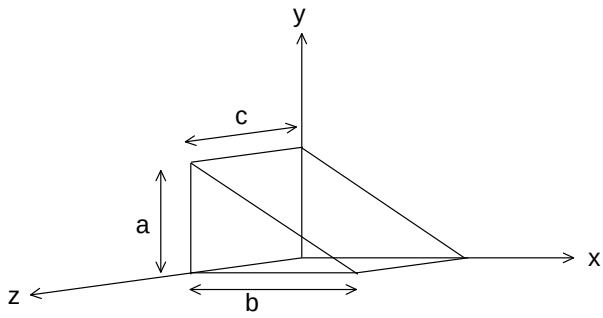


Space for rough work

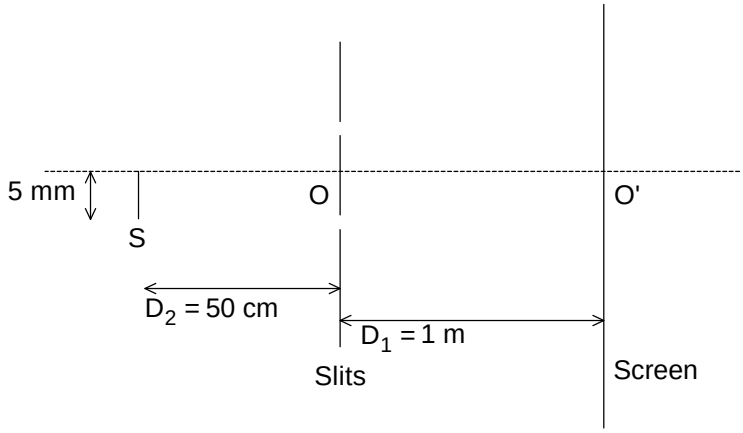
5. A block rest on a long plank moving at 2 ms^{-1}. The co-efficient of friction between the block and plank is 0.1. The plank starts decelerating uniformly and stops in 0.5 s. Acceleration due to gravity is 10 ms^{-2}. The distance slid by block on plank is $\frac{K}{4}$ m. Find K …



6. In the given electric field $\vec{E}=\left[(d+x)i+E_0\hat{j}\right]$ N/C a hypothetical closed surface is taken as shown in figure. The total charge enclosed within the closed surface is $k\in_0$. Then find the value of k(a = 2 m, b = 3 m and c = 2 m)



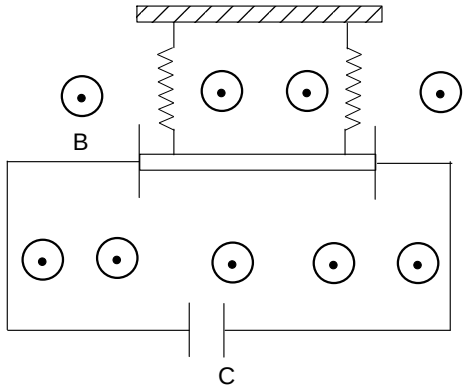
7. In the given Y.D.S.E the distance between the slits = 1 cm and the distance between slits and screen is D_1 = 1 m. If the source of the light S is placed at distance below 5 mm below the line OO′ at distance D_2 = 50 cm from slits. Find distance of central maxima from O′ (in cm)?



Space for rough work

8. A gas of hydrogen like atoms can absorb radiation of 68 eV in first excited state. Consequently the gas emit radiations of only three different wavelengths. All the wavelengths are equal or smaller than that of the absorbed photon. The atomic number of gas atom is …

9. A conducting rod of mass m and length ℓ is suspended horizontal with help of two identical non-conducting springs each of stiffness K in presence of uniform horizontal magnetic field of induction B. The rod can slid between two fixed frictionless vertical conducting guides as shown in figure. The rod is pulled slightly downward and then released. Neglect resistance and inductance in circuit. Acceleration due to gravity is g. The time period of oscillation is $T = k'.\pi$. If $m = 7B^2\ell^2C$ and m = k. Find k′ ?



10. You have m = 1 kg of super cooled water at temperature – 10°C kept in a container and crushed at temp θ = - 20°C kept in another container. How much ice you should add to the water so that whole water freezes $S_\omega = 4.2\times10^3\ J\ kg^{-1}K^{-1}$, $S_l = 2.1\times10^3\ J\ Kg^{-1}K^{-1}$. Latent of heat of ice $L = 336\times10^3\ J\ Kg^{-1}$ and meeting point of ice is θ = 0°C. Heat capacity of the vessel and heat less to surrounding is negligible.

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

1. Select correct statement(s).
   (A) all N – N bond lengths are same in N_3^- ion
   (B) all N – N bond lengths are not identical in N_3H
   (C) in HN_3 terminal N – N bond length is shorter than the central N – N bond length
   (D) azide ion and hydrazoic acid have same number of electron.

2. For the Br^- catalyzed aqueous reaction
   $HNO_2 + C_6H_5NH_2 + H^+ \xrightarrow{Br^-} C_6H_5N_2^+ + 2H_2O$
   proposed mechanism is
   (I) $H^+ + HNO_2 \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} H_2NO_2^+ \ (fast)$
   (II) $H_2NO_2^+ + Br^- \xrightarrow{K_2} ONBr + H_2O \ (slow)$
   (III) $ONBr + C_6H_5NH_2 \xrightarrow{K_3} C_6H_5N_2^+ + H_2O + Br^- \ (fast)$
   Which of the following statement(s) is/are correct?
   (A) order with respect to H^+ is 1
   (B) order with respect to Br^- is 1
   (C) order with respect to $C_6H_5NH_2$ is 1
   (D) order with respect to HNO_2 is 1

3. The order of first ionization energies are given below. Find the incorrect order(s).
   (A) $F > N > O > C > B > Be$
   (B) $C > Si > Ge > Pb > Sn$
   (C) $B > Al > Ga > In > Tl$
   (D) $O > Cl > Br > I > S$

4. In antifluorite structure
   (A) oxide ions have face centred cubic arrangement
   (B) cations are present in all the tetrahedral voids
   (C) $r_{Na^+} + r_{O^{2-}} = \frac{\sqrt{3}\,a}{4}$
   (D) cations are face centred and anions are present in all the tetrahedral voids

Space for rough work

5. Identify the incorrect statement(s).

(A) $[NiCl_4]^{2-}$ as well as $[Ni(CO)_4]$ are paramagnetic

(B) CFSE for $[Ti(H_2O)_6]^{3+}$ is $-0.4\Delta_0$

(C) Δ_0 increasing order is $[CrCl_6]^{3-} < [Cr(NH_3)_6]^{3+} < [Cr(CN)_6]^{3-}$

(D) in spectro chemical series, halide ions are arranged as $I^- < Br^- < Cl^- < F^-$

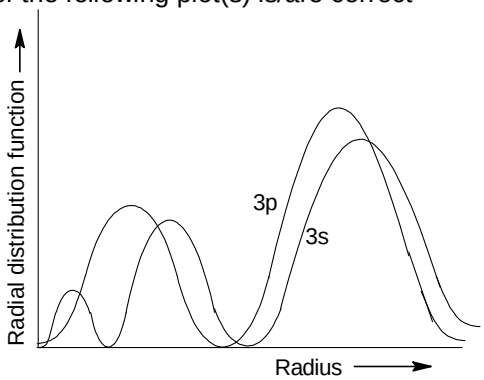
6. On being treated with HCl, acetone gives

(A) mesityl oxide (B) phorone

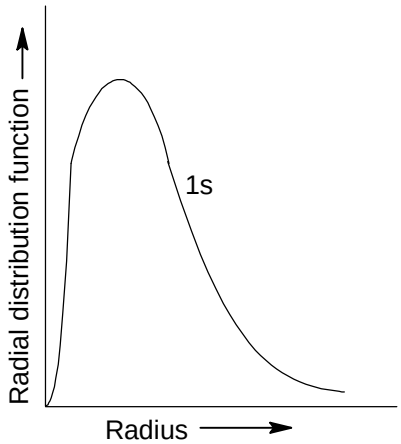
(C) mesitylene (D) aldol

7. Which of the following plot(s) is/are correct

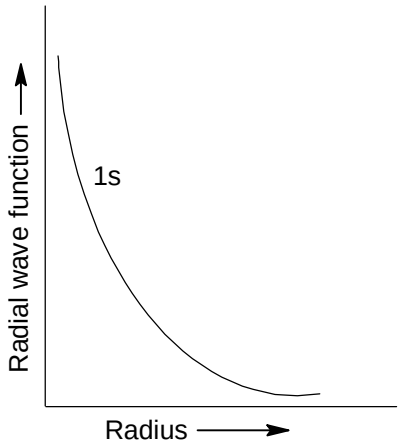
(A)



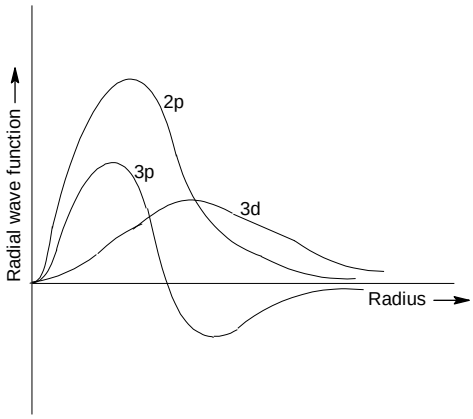
(B)



(C)



(D)



Space for rough work

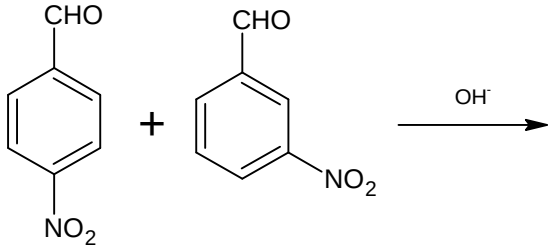
8. $CH_3 - CH_2 - CH_2 - NH_2 \xrightarrow{N_2O_3}$

The product(s) formed is/are

(A) $H_3C-\underset{\underset{OH}{|}}{CH}-CH_3$ (B) $CH_3 - CH_2 - CH_2 - OH$

(C) $CH_3 - CH = CH_2$ (D) $CH_3 - CH_2 - C \equiv N$

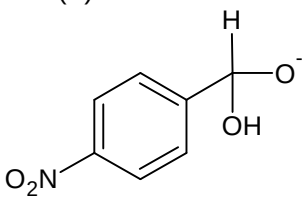
9. Which of the following statement(s) is/are correct for crossed cannizzaro reaction?

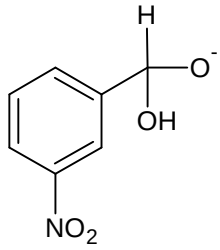


(A) (B)

(A) OH^- attacks at the carbonyl C – atom of (A)

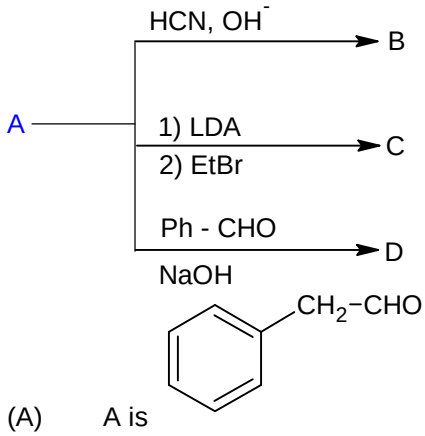
(B) OH^- attacks at the carbonyl C – atom of (B)

(C) the H^- ion transfer takes place from 

(D) the H^- ion transfer takes place from 

Space for rough work

10. $C_8H_8O(A) \xrightarrow{I_2, NaOH}$ yellow ppt.



(A) A is

(B) B is optically active

(C) C is $Ph-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-CH_3$

(D) D is $Ph-CH=CH-\overset{O}{\overset{\|}{C}}-Ph$

SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. $H_3C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=CH_2 \xrightarrow{\text{Reagent}}$ Alcohol

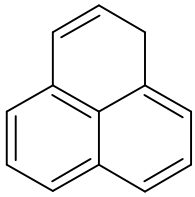
Following are the possibilities of products on applying specific reagents:

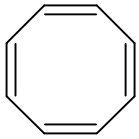
	Reagent	Possible product
1.	B_2H_6 / H_2O_2, $NaOH$	$H_3C-\underset{\displaystyle CH_3}{\underset{\|}{CH}}-CH_2-CH_2-OH$
2.	$PdCl_2$, H_2O, O_2 / LAH	$H_3C-\underset{\displaystyle CH_3}{\underset{\|}{CH}}-\underset{\displaystyle HO}{\underset{\|}{CH}}-CH_3$
3.	$Hg(OAc)_2$, H_2O / $NaBH_4$	$H_3C-\underset{\displaystyle H_3C}{\underset{\|}{\overset{\displaystyle OH}{\overset{\|}{C}}}}-CH_2-CH_3$
5.	dil. H_2SO_4	$H_3C-\underset{\displaystyle H_3C}{\underset{\|}{\overset{\displaystyle OH}{\overset{\|}{C}}}}-CH_2-CH_3$

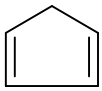
How many of the above mentioned pairs are correct?

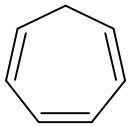
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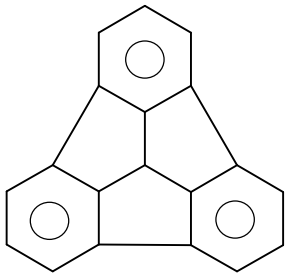
2. How many of the following reactions give aromatic species?

(I)  + KH ⟶

(II)  + 2K ⟶

(III)  $\xrightarrow{NaNH_2}$

(IV)  $\xrightarrow{NaNH_2}$

(V)  $\xrightarrow{NaNH_2}$

(VI)  $\xrightarrow{NaNH_2}$

3. The formation constant is $x \times 10^5$ for the reaction of a tripositive metal ion with thiocyanate ion to form the mono complex. If the total metal concentration in the solution is 2.0×10^{-3} M, the total SCN^- concentration is 1.50×10^{-3} M, and the free SCN^- concentration is 1.0×10^{-5} M. Find the value of x.

4. One of the most stable nuclei is ^{55}Mn. Its nuclidic mass is 54.9244 u. Determine its average binding energy per nucleon in MeV if mass of proton and neutron are 1.00783 u and 1.00867 u respectively.

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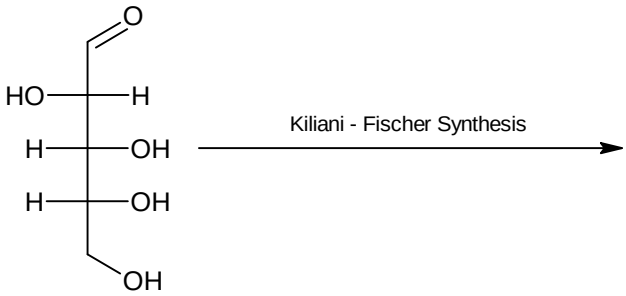
5. How many of the following chemical compound(s) appear black?
MnO_2, SnS, CO_3O_4, Ni_2O_3, $K_2Cr_2O_7$, CoS, anhydrous $CoSO_4$, Bi_2S_3, HgS, Sb_2S_3

6. K_a for the acid ionization of Fe^{3+} to $Fe(OH)^{2+}$ and H^+ is 5.26×10^{-7}. What is the maximum pOH value which could be used so that at least 95% of the total Iron(III) in a dilute solution exists as Fe^{3+}?

7. How many of the following chemical reagents impart brick red colouration to the flame during flame test? Dolomite, bleaching powder, rock salt, nitrolim, lime-stone, carnallite, alum, gypsum.

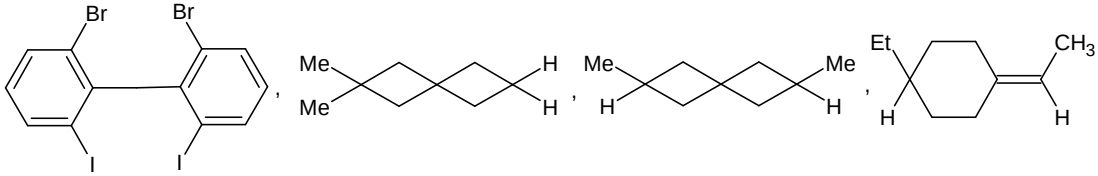
8. The number of P – O – P bonds in cyclic trimeric metaphosphoric acid are

9. How many products are formed for the following synthesis?



D-Arabinose

10. How many of these show optical activity?



$CH_3 - CH = C = C = CH - C_2H_5$, $CH_3 - CH = C = C = C = CH - C_2H_5$, cis-cyclooctene, trans-cyclooctene, (2R, 3S) 2, 3-dihydroxybutane-1, 4-dioic acid.

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

1. Let $f(n)=\sum_{K=n+1}^{2n}(K-n)^2\left({}^nC_{K-n}\right)^2$ where $n \geq 2$ then

(A) $f(n)=n^2\,{}^{2n}C_n-2n$

(B) $f(n)=\left(\frac{n^3+131n-150}{30}\right){}^{2n-2}C_{n-1}$

(C) $f(25)=\frac{625\times 48!}{(24!)^2}$

(D) $f(25)=\frac{625\times 50!}{(25!)^2}-50$

2. The probability that a zoo has exactly n tiger is P^n, $n \geq 1$. All sex distribution of n tiger in that zoo have the same probability then the probability that the zoo contain exactly K male tiger (where $K \geq 1$) is

(A) $\frac{1}{1-P}\left(\frac{P}{1-P}\right)^{K-1}$

(B) $\frac{2}{1-P}\left(\frac{P}{1-P}\right)^{K}$

(C) $\left(\frac{2}{2-P}\right)\left(\frac{P}{1-P}\right)^{K-1}$

(D) none of these

3. Which holds true if A is a square matrix of order n.

(A) $\text{adj adj}\,(\text{adj } A)=|A|^{n^2-3n+2}A^{-1}$

(B) $\text{adj adj}\,(\text{adj } A)=|A|^{n^2-3n+3}A^{-1}$

(C) $\left|A^{-1}\text{adj adj}\,(\text{adj}A)\right|=|A|^{(n-1)^3+1}$

(D) $\left|A^{-1}\text{adj adj}\,(\text{adj}A)\right|=|A|^{(n-1)^3-1}$

4. The equation of the straight line which meets the circle $x^2+y^2-2x-4y-p^2=0$ at points which are at a distance 'd' from a point $A(1+\alpha, 2+\beta)$ on the circle is

(A) $2\alpha x+2\beta y-2(\alpha+2\beta)=2(p^2+5)-d^2$

(B) $2\alpha x-2\beta y-2(\alpha-2\beta)=2(p^2+5)+d^2$

(C) $2\alpha x+2\beta y-2(\alpha+2\beta)=2(p^2+5)+d^2$

(D) $2\alpha x+2\beta y-2(\alpha+2\beta)=d^2-2(p^2+5)$

Space for rough work

5. Let S & S′ be the foci of an ellipse $b^2(x-2)^2 + a^2(y-1)^2 = a^2b^2$ whose eccentricity is e and P is a variable point on the ellipse then
(A) locus of incentre of $\Delta PSS'$ is hyperbola
(B) locus of incentre of $\Delta PSS'$ is ellipse
(C) eccentricity of the locus of incentre of $\Delta PSS'$ is $\sqrt{\frac{2e}{e+1}}$
(D) eccentricity of the locus of incentre of $\Delta PSS'$ is $\sqrt{\frac{2e}{e-1}}$

6. Let f be a function satisfying $f_e(x) = \frac{e^x}{e^x + \sqrt{e}}$ and $g(x) = f_4(x)$
(A) $f_2(x) + f_2(1-x) = 1$
(B) $\sum_{r=1}^{1996} g\left(\frac{r}{1997}\right) = 1996$
(C) $\sum_{r=1}^{2n-1} 2f\left(\frac{r}{2n}\right) = (2n-1)$
(D) all of the above

7. Two tangents on a parabola are x – y – 1 = 0 & x + y + 11 = 0. If S(1, 3) is focus of the parabola where P & Q are end points of focal chord then
(A) equation of tangent at vertex 3y – 2x + 1 = 0
(B) equation of directrix 3y – 2x + 8 = 0
(C) latus rectum is $\frac{30}{\sqrt{13}}$
(D) $\frac{1}{SP} + \frac{1}{SQ} = \frac{2\sqrt{13}}{15}$

8. If a, b, c, d are in GP where a, b, c, d are positive number then
(A) a – b > d – c
(B) ab + cd > 2(ac + bd + bc)
(C) $2\left(c^{-1}d^{-1} + a^{-1}b^{-1}\right) > 2\left(b^{-1}d^{-1} + a^{-1}c^{-1} - a^{-1}d^{-1}\right)$
(D) $\frac{1}{d} + \frac{1}{b} > \frac{2}{c}$

Space for rough work

9. $\lim_{x\to\infty} x_{c_n}\left(\frac{a}{x}\right)^n\left(1-\frac{a}{x}\right)^{x-n}$ is equal to

(A) $\frac{n!}{a^n e^{-a}}$ (B) $\frac{a^n e^{-a}}{n!}$

(C) $\frac{n!}{a^n e^{a}}$ (D) $\frac{a^n e^{a}}{n!}$

10. Suppose we uniformly and randomly select a permutation from the 20! Permutation of 1, 2, 3 …. 20. What is the probability that 2 appears at an earlier position than any other even number in the selected permutation.

(A) $\frac{1}{2}$ (B) $\frac{1}{10}$

(C) $\frac{9!}{20!}$ (D) none of these

SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A quadrilateral ABCD in which side AB, BC, CD, DA are in GP such that circle can be inscribed in it and another circle is circumscribed about it, then angle A is equal to $\frac{n\pi}{q}$ where n & q are coprime then (n + q) equal to

2. Set $S=\{|y|=|x|, |x|-r=0=|y|-r \,/\, r\in N, 1\le r\le n\}$ then total number of triangles formed by the lines in the given sets is K(2n) then K is …

3. In ΔABC, ΔADB, ΔADC median AD, AE, AF respectively is drawn and BC is equal to 4, then the value of $AE^2 + AF^2 - 2AD^2$ is

4. The volume of the reel – shaped solid formed by the revolution about the y-axis of the part of the parabola $y^2 = 2x$ cut off by the latus rectum is P then value of $\frac{20}{\pi}P$ is

Space for rough work

5. If a + 1 > b and A, B, C are variable angles such that $\sqrt{\left(a^2-1\right)+b\left(2-b\right)}\tan A+a\tan B+\sqrt{\left(a^2+1\right)+b\left(b-2\right)}\tan C=3a$ then value of $\sum\tan^2 A$ is

6. Let PQRSTU is a hexagon such that PQ = QR = RS = 3 and ST = TU = UP = 6. If the vertices P, Q, R, S, T, U are concyclic, then the radius of the circle passing through them is $\sqrt{a}$, then $\frac{a}{3}$ is

7. If $\int_0^{\pi}\left[\frac{\cos(n+1)x-\cos nx}{\cos x-1}\right]dx=\frac{n\pi}{2}$ then value of n is

8. Assuming x – y + 2z = 0 to be the horizontal plane then equation of line of greatest slope through the point (2, 2, 1) in the plane 2x + y – 6z = 0 is L having cosine ratio a : b : c then value of $\left(\frac{2a+b}{c}\right)$ is ..

9. Number of real value of x satisfy this equation $e^{2x}+e^{x}\left(3-5\cos x\right)+1=0$

10. If $\sum_{i=1}^{45}\left[\sin^{-1}\left\{\mathrm{Re}\left(\frac{ai+bi\omega+ci\omega^2}{bi+ai\omega^2+ci\omega}\right)\right\}+\cos^{-1}\left\{\mathrm{Re}\left(\frac{ai+bi\omega+ci\omega^2}{bi\omega^2+ci+ai\omega}\right)\right\}\right]$ is equal $\frac{90\pi}{x}$ then x equal to

Space for rough work